

Gallbladder and biliary tree

The biliary tree consists of the system of ducts that collect and deliver bile from the liver to the second part of the duodenum. It is conventionally divided into intrahepatic and extrahepatic biliary trees. The intrahepatic ducts are formed from bile ductules that join to form segmental ducts. These merge to form right and left hepatic ducts close to the porta hepatis. The extrahepatic biliary tree consists of the extrahepatic segments of the right and left hepatic ducts, the common hepatic duct, the cystic duct and gallbladder, and the common bile duct (Fig. 68.1).

GALLBLADDER

The gallbladder is a flask-shaped, blind-ending diverticulum attached to the bile duct by the cystic duct (Fig. 68.2). It stores and concentrates bile. In life, it is grey-blue in colour and is usually firmly attached by connective tissue to the inferior surface of the right lobe of the liver, between segments IV and V at the lower limit of the principal plane. In the adult, the gallbladder is between 7 and 10 cm long, with a resting volume of about 25 ml and a capacity of up to 50 ml (Di Ciaula et al 2012). It usually lies in a shallow fossa (the gallbladder bed) on the visceral surface of the right lobe of the liver, covered by peritoneum continued from the liver surface. This attachment can vary widely. Rarely, the gallbladder is almost completely buried within the liver (intrahepatic gallbladder; Guiteau et al 2009), or suspended from the liver by a peritoneal mesentery (when it is at risk of torsion; Gupta et al 2009), or connected to the duodenum by an extension of the free edge of the lesser omentum (cystoduodenal ligament; Ashaolu et al 2011).

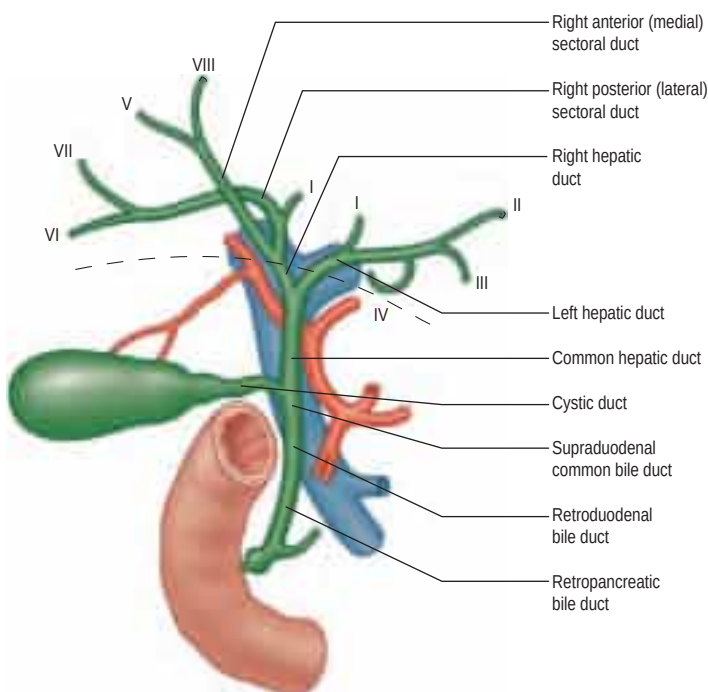


Fig. 68.1 The overall arrangement of the intrahepatic and extrahepatic biliary tree. The segmental ducts often branch just before they enter the main ducts, or are multiple as they enter the main ducts, but for clarity are shown here as single ducts. Note that segment I often drains via both right and left hepatic ducts. The level of the liver parenchyma at the porta hepatis is shown by the dashed line.

The gallbladder is described as having a fundus, body and neck. The neck lies at the medial end, close to the porta hepatis, and almost always has a short peritoneal attachment (mesentery) to the liver, which usually contains the cystic artery. The mucosa at the medial end of the neck is obliquely ridged, forming a crescentic fold that is continuous with the spirally arranged mucosal folds in the cystic duct (Dasgupta and Stringer 2005). At its lateral end, the neck widens out to form the body of the gallbladder; when this widening is clearly demarcated as a result of gallstone disease, it is referred to as 'Hartmann's pouch'. The neck usually lies anterior to the second part of the duodenum. The body of the gallbladder normally lies in contact with the visceral surface of the liver. When the neck possesses a mesentery, this rapidly shortens along the length of the body as it lies in the gallbladder fossa. The body lies anterior to the second part of the duodenum and the right end of the transverse colon. The bulbous fundus lies at the lateral end of the body and usually projects past the inferior border of the liver to a variable extent. Here, it frequently lies in contact with the anterior abdominal wall behind the ninth costal cartilage, where the lateral edge of the right rectus abdominis crosses the costal margin. This is where enlargement of the gallbladder is best sought on clinical examination. The fundus commonly lies adjacent to the transverse colon.

The gallbladder varies in size and shape. The fundus may be elongated and highly mobile. Rarely, the fundus is folded back on the body of the gallbladder, the so-called Phrygian cap; on ultrasound, this may be wrongly interpreted as an apparent septum within an otherwise normal gallbladder. Other anatomical variants of the gallbladder include duplication, with or without a double cystic duct; agenesis; internal septation; and an ectopic location (most commonly left-sided) (Gross 1936, Faure et al 2008); although rare, these congenital anomalies are particularly important if the patient requires surgery for gallbladder or gallstone disease (Lamah et al 2001, Singh et al 2006, Castorina et al 2014, Chowbey et al 2004).

INTRAHEPATIC BILIARY TREE

SEGMENTAL AND SECTORAL DUCTS

The segmental ducts of the left hemi-liver have a relatively constant pattern, although more than one segmental duct may drain each particular segment. The left hepatic duct is formed by the union of segment II and III ducts, most often behind or to the left of the umbilical portion of the left portal vein (see Fig. 68.1). The biliary drainage of segment IV is more variable but is usually by a single duct into the left hepatic duct. The right hepatic duct is formed by the union of the right anterior (medial) and posterior (lateral) sectoral ducts. The right anterior (medial) sectoral bile duct drains segments V and VIII, and the right posterior (lateral) sectoral duct drains segments VI and VII. The right posterior sectoral duct usually curves around the posterior aspect of the right anterior duct before fusing with its medial aspect; this is known as Hjortsjö's crook and is an important technical consideration when performing liver resection (Fig. 68.3). The bile ducts draining the caudate lobe (segment I) usually join the origin of the left hepatic duct or may drain into both hepatic ducts near the hilar confluence.

The right hepatic duct and its branches are more often subject to variation than the left ductal system (Cucchetti et al 2011, Chaib et al 2014). These variations have been classified into six main types (Table 68.1, Fig. 68.4). Left intrahepatic ductal variations mostly relate to the drainage pattern of segment IV; this segmental bile duct usually drains into the left hepatic duct but it may open into a segment II or III bile duct, the right anterior sectoral duct, or even the common hepatic duct (Chaib et al 2014).

An oblique groove on the inferior surface of the liver posterior to the gallbladder bed is present in 70–80% of livers. It is variably known as the fissure of Gans, Rouviere's sulcus or the incisura hepatis dextra ([Ch. 67](#)). It overlies the division of the right posterior (lateral) portal pedicle, where it gives off the inferior segment VI branch. It has been increasingly recognized as a useful anatomical landmark during hepatic resection and in laparoscopic cholecystectomy (since the cystic duct and artery lie anterosuperior to the sulcus while the common bile duct lies posteroinferior) ([Dahmane et al 2013](#)).

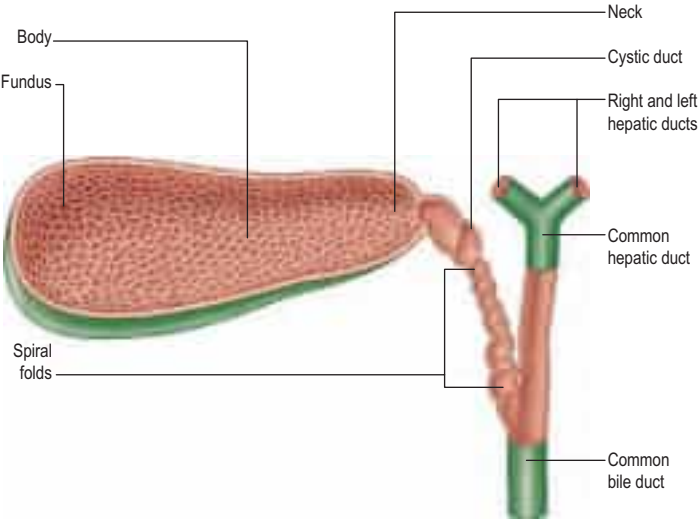


Fig. 68.2 The interior of the gallbladder and bile ducts.

Table 68.1 Major variations of the intrahepatic duct drainage patterns

Type	Approximate % of population	Description
1	60*	Normal anatomy
2	15	No right hepatic duct. The common hepatic duct is formed by the union of right anterior sectoral, right posterior sectoral and left hepatic ducts (trifurcation pattern)
3	15	One of the right sectoral ducts (more often the posterior) joins the left hepatic duct
4	10	Low drainage of one of the right sectoral ducts (more often the posterior) into the common hepatic duct
Rare	<5	The segment V duct or the right posterior sectoral duct drains into the cystic duct or gallbladder

(Data from Cucchetti et al 2011 and Chaib et al 2014.)
*This figure is higher in men and lower in women, i.e. women are more likely to have variant intrahepatic bile duct anatomy. N.B. The percentage of different types varies between populations.

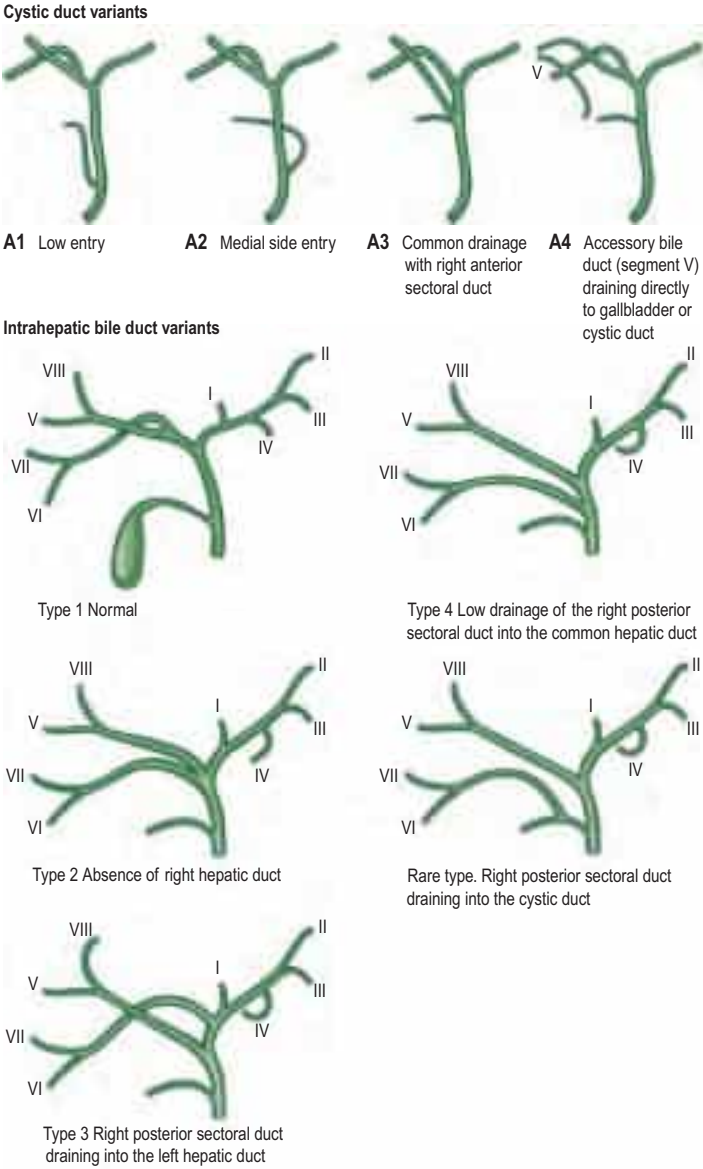


Fig. 68.4 Variations in the anatomy of the cystic and intrahepatic bile ducts. The cystic duct variations are labelled A1–A4 and the intrahepatic bile duct variations as types 1–4 (see Table 68.1).

EXTRAHEPATIC BILIARY TREE

CYSTIC DUCT

The cystic duct drains the gallbladder into the common bile duct. In adults, it is usually between 2 and 4 cm long and has a luminal diameter of 2–3 mm (Dasgupta and Stringer 2005). It passes posteriorly and medially from the neck of the gallbladder, often in a tortuous fashion, to unite with the common hepatic duct and form the common bile duct. The anatomy of the junction between the cystic duct and common hepatic duct is variable (see Fig. 68.4). In most individuals, the cystic duct joins the middle third of the combined lengths of the common hepatic and common bile ducts (Shaw et al 1993), but it may drain into the distal common bile duct or into a more proximal duct such as the proximal common hepatic duct or right hepatic duct; it usually joins the right lateral aspect of the common hepatic duct but may merge medially, anteriorly or posteriorly; and it usually forms an oblique angle with the common hepatic duct but can spiral around it or run parallel to it in the free edge of the lesser omentum for a variable distance before merging (Lamah et al 2001). Irrespective of the site or type of union, the terminal part of the cystic duct is frequently adherent to the common hepatic duct for a variable distance.

Rarely, the cystic duct is double or absent (when the gallbladder drains directly into the bile duct), or receives an anomalous hepatic duct from segment V of the liver. These variations in cystic duct anatomy are of considerable importance during surgical excision of the gallbladder (cholecystectomy). The cystic duct must be identified passing to the neck of the gallbladder and must be occluded some distance away from the bile duct to prevent injury to the latter. A preliminary operative cholangiogram is essential if the anatomy is unclear or anomalous.

The mucosa of the cystic duct has 2–10 crescentic folds that project into the lumen and form a spiral; these are continuous with those in the neck of the gallbladder. The function of these spiral folds is unknown but they may help to preserve the patency of this narrow, tortuous duct

rather than regulate the flow of bile, as is commonly stated (Dasgupta and Stringer 2005).

HEPATIC DUCTS

The right and left hepatic ducts emerge from the liver and unite near the right end of the porta hepatis to form the common hepatic duct. The extrahepatic right duct is short (0.5–2.0 cm in adults) and nearly vertical, while the left is longer (1.5–3.5 cm) and more horizontal, and lies along the inferior border of segment IV. The accessibility of the extrahepatic segment of the left hepatic duct is exploited when performing a surgical biliary bypass in patients with benign hilar bile duct strictures (Myburgh 1993).

In adults, the common hepatic duct descends approximately 3 cm before being joined obliquely on its right by the cystic duct to form the common bile duct. The common hepatic duct lies to the right of the hepatic artery and anterior to the portal vein in the free edge of the lesser omentum. In adults, the luminal diameter of the normal common hepatic duct, as measured by ultrasound, is less than 5 mm.

COMMON BILE DUCT

The common bile duct is formed near the porta hepatis, by the junction of the cystic and common hepatic ducts (Figs 68.5–68.6). In adults, it is usually between 6 and 8 cm long and its luminal diameter, as

A small bile duct from segment V of the liver may traverse the gallbladder fossa and join the right hepatic duct or its anterior sectoral branch or the common hepatic duct. This is often known as Luschka's duct and its importance lies in the fact that it may be injured during cholecystectomy, causing a postoperative bile leak (Spanos and Syrakos 2006). Postmortem studies have identified such a duct in up to one-third of individuals but many of these are small and relatively insignificant; if larger ducts (1–2 mm) are considered, the prevalence is nearer 5% (Ko et al 2006). These ducts are more likely to be injured if the gallbladder is not dissected close to its wall. Luschka did not describe a duct draining directly into the gallbladder; this is referred to as a cystohepatic or cholecystohepatic duct and should not be confused with a hepatocystic duct (which is present when the common bile duct is absent and both hepatic ducts drain directly into the gallbladder which drains in turn via the cystic duct into the duodenum) (Losanoff et al 2002).

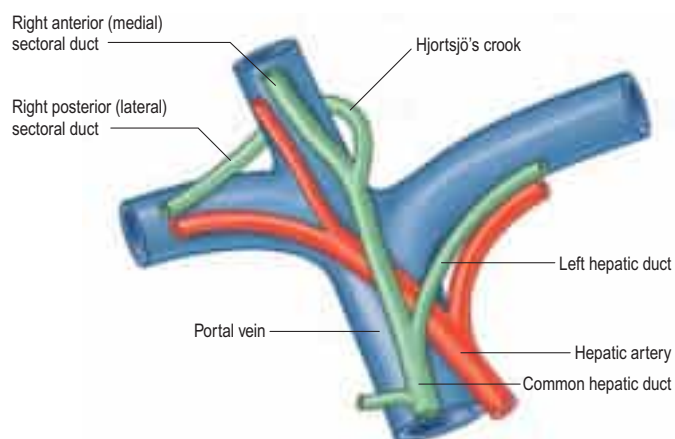


Fig. 68.3 Hjortsjö's crook. The right posterior (lateral) sectoral bile duct usually hooks around the right anterior (medial) sectoral pedicle, rendering it vulnerable to injury when performing an extended left hepatectomy that includes segments V and VIII of the liver (left trisectionectomy).



Fig. 68.5 A, An endoscopic retrograde cholangiopancreatogram. (B, continued online)



Fig. 68.6 A magnetic resonance cholangiopancreatogram.

measured by ultrasound, is no more than 7 mm (Perret et al 2000); the diameter increases slightly with advancing age from a mean of 3.6 mm below 60 years to a mean of 4 mm after 80 years. The common bile duct can be divided into supraduodenal, retroduodenal and pancreatic segments. The supraduodenal segment descends posteriorly and slightly to the left, anterior to the epiploic foramen and inferior vena cava, in the free right border of the lesser omentum, where it lies anterior and to the right of the portal vein and to the right of the hepatic artery. This part of the common bile duct is the most accessible at surgery. The retroduodenal segment lies behind the first part of the duodenum with the gastroduodenal artery on its left. The pancreatic segment runs in a groove on the posterior surface of the head of the pancreas, embedded in the gland to a variable degree; it lies up to 2 cm away from the medial

wall of the second part of the duodenum and anterior to the right renal vein. The posterior superior pancreaticoduodenal branch of the gastroduodenal artery descends anterior to the retroduodenal portion of the common bile duct (at the superior border of the pancreas) before spiralling around the right side of the bile duct to reach the posterior surface of the head of the pancreas (Bertelli et al 1996).

Hepatopancreatic ampulla

As it descends behind the head of the pancreas medial to the second part of the duodenum, the common bile duct approaches the right end of the pancreatic duct. The two ducts usually enter the duodenal wall together in a Y configuration to form a short common channel measuring between 2 and 10 mm in length (Flati et al 1994). This common channel often contains a dilation known as the hepatopancreatic ampulla (of Vater) and it opens via a single orifice on to the medial wall of the second part of the duodenum at the major duodenal papilla (Fig. 68.5B). In clinical practice, the whole region is often termed the pancreaticobiliary junction. Occasionally, the common bile duct and pancreatic duct unite outside the duodenal wall to form an abnormally long common channel or the two ducts are separated by a septum or drain into the duodenum separately (Kamisawa and Okamoto 2008).

The mucosa of the terminal 5–10 mm of the common bile and pancreatic ducts has a complex arrangement of circumferentially arranged folds (Purvis et al 2013) (Fig. 68.7). Their distribution and orientation impede reflux of duodenal contents into the bile and pancreatic ducts, and they may sometimes cause difficulty cannulating the major duodenal papilla during endoscopic retrograde cholangiopancreatography (ERCP).

A complex arrangement of smooth muscle with prominent circular fibres surrounds the distal common bile duct (bile duct sphincter) and hepatopancreatic ampulla (sphincter of Oddi); to a lesser extent, it also surrounds the terminal part of the main pancreatic duct (pancreatic duct sphincter) (Boyden 1957, Didio and Anderson 1968). The whole sphincter muscle complex lies mostly within the wall of the duodenum and, in adults, measures approximately 15–20 mm in length; it contains a high pressure zone that can be detected by manometry (Teilmann 1991, Habib et al 1988). The smooth muscle of the ampullary sphincter is developmentally and anatomically distinct from the adjacent duodenal muscle. The sphincter regulates the flow of bile and pancreatic secretions into the duodenum and impedes reflux of duodenal contents into the ductal system (Guelrud et al 1990); it is inhibited by cholecystokinin (CCK), which not only relaxes the sphincter but also causes gallbladder contraction (Woods et al 2005). Division of the upper part of the ampullary sphincter (sphincterotomy) may be required to allow access to the common bile duct during ERCP.

The pancreaticobiliary junction is clinically important because it may be affected by various congenital and acquired disorders. An anomalous union between the bile and pancreatic ducts, particularly one resulting in an abnormally long common channel, may be associated with congenital bile duct dilation, recurrent pancreatitis, and/or gallbladder cancer (Kimura et al 1985, Misra and Dwivedi 1990, Stringer et al 1995). Acquired pathology in this region also includes gallstone obstruction and peri-ampullary tumours.

Hepatobiliary triangle

The triangular region formed between the cystic duct, the common hepatic duct and the inferior surface of the liver is the hepatobiliary triangle (Fig. 68.8). It is often mistakenly referred to as Calot's triangle, which is an isosceles triangle based on the common hepatic duct, with the cystic artery and cystic duct forming its sides (Stringer 2009). The hepatobiliary triangle is bridged by the double layer of peritoneum that forms the short and variable mesentery of the cystic duct. Between these two layers there is a variable amount of fatty connective tissue, lymphatics, the cystic lymph node, autonomic nerves, and usually the cystic artery as it runs from the right hepatic artery to the gallbladder; occasionally, there may also be an accessory bile duct (see above). Understanding the variations in biliary and arterial anatomy as they relate to the triangle is of considerable importance during excision of the gallbladder in order to avoid injury to the common hepatic or common bile duct or right hepatic artery (Suzuki et al 2000, Talpur et al 2010).

GALLSTONES

Gallstones are relatively common and usually form in the gallbladder. Gallbladder contraction may move a gallstone towards the narrow cystic



Fig. 68.5 B, Endoscopic image of the major duodenal papilla (arrow) on the medial wall of the descending duodenum.

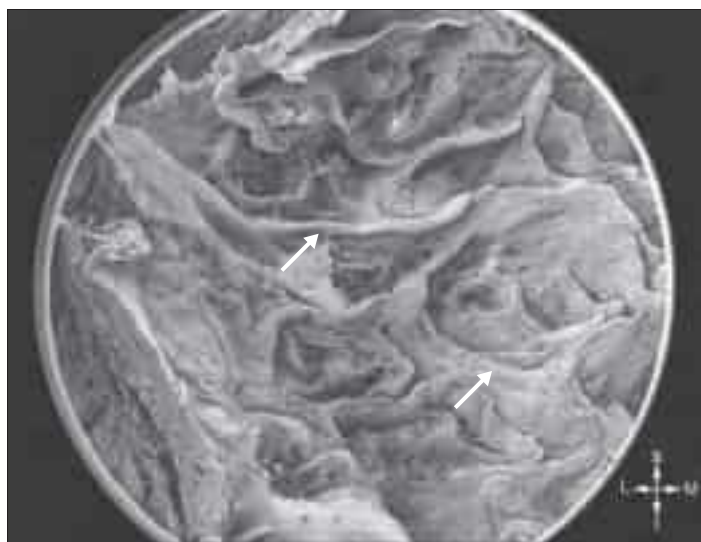


Fig. 68.7 A scanning electron micrograph showing mucosal folds (arrows) lining the wall of the hepatopancreatic ampulla. Abbreviations: I, inferior; L, lateral; M, medial; S, superior.

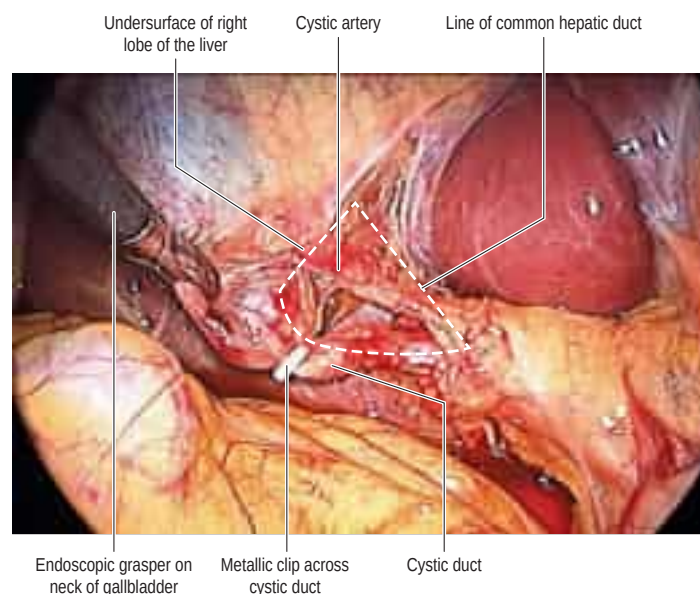


Fig. 68.8 A view of the hepatobiliary triangle (dashed line) at laparoscopic cholecystectomy. In this patient, the cystic artery can be seen crossing the triangle superficially *en route* to the gallbladder. (Courtesy of Richard Flint, Christchurch Hospital.)

duct, where it can cause obstruction and pain (biliary colic). If the stone continues to occlude the cystic duct, it will lead to inflammation of the gallbladder (acute cholecystitis) or a sterile distension of the gallbladder (mucocele); providing the gallbladder has not been inflamed previously, it will be non-fibrotic and distensible, and the fundus often becomes palpable below the costal margin. A gallstone lodged in the cystic duct or neck of the gallbladder may also produce an inflammatory swelling that compresses the common hepatic duct, resulting in partial obstruction to the flow of bile and the development of jaundice (the so-called 'Mirizzi syndrome'; [Stringer 2009](#)). If the gallstone passes through the cystic duct, it may become impacted in the distal common bile duct or at the pancreaticobiliary junction, causing obstructive jaundice, cholangitis and/or acute pancreatitis.

ENDOSCOPIC CHOLANGIOPANCREATOGRAPHY

The common bile duct and/or pancreatic duct may be accessed endoscopically from the duodenum for diagnostic cholangiography and/or pancreatography, and for therapeutic interventions such as the relief of biliary obstruction by insertion of an internal stent. The acute angle between the distal common bile duct and pancreatic duct may make direct cannulation of the bile duct difficult. In such cases, the upper part of the major duodenal papilla may be incised (pre-cut sphincterotomy) to facilitate cannulation of the common bile duct. This incision, and the more extensive sphincterotomy that may be performed to release an impacted gallstone ([Sakai et al 2012](#)), are occasionally complicated by arterial bleeding. The risk may be reduced by making the incision in the 10–11 o'clock region of the sphincter ([Mirjalili and Stringer 2011](#)). Endoscopic sphincterotomy can result in uncontrolled reflux of duodenal contents into the distal common bile duct and ascending biliary infection.

BILIARY DRAINAGE

The proximity of the fundus of the gallbladder to the anterior abdominal wall facilitates ultrasound-guided percutaneous drainage of a distended, obstructed gallbladder. Because of the small calibre of the cystic duct and its spiral folds, drainage of the gallbladder is rarely sufficient to decompress the biliary tree when the common bile duct is obstructed. In such cases, temporary or permanent biliary drainage may be achieved by the endoscopic insertion of a biliary stent through the obstruction via the major duodenal papilla, surgical bile diversion, or percutaneous transhepatic drainage. The latter involves image-guided puncture of a dilated intrahepatic bile duct and the insertion of an external drain or an internal stent through the biliary obstruction.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Cystic artery

The cystic artery usually arises from the right branch of the hepatic artery ([Fig. 68.9](#), but see [Fig. 68.10](#)) and most often passes posterior to

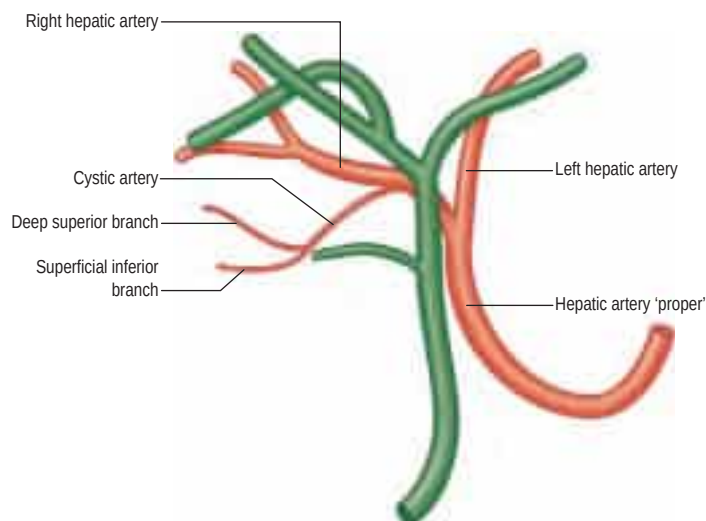


Fig. 68.9 Normal anatomy of the cystic artery.

the common hepatic duct within the hepatobiliary triangle, where it lies superior to the cystic duct. On reaching the superior aspect of the neck of the gallbladder, it divides into a superficial branch that runs along the peritoneal (inferior) surface of the gallbladder, and a deep branch that runs between the gallbladder and its fossa; the branches anastomose over the surface of the body and fundus. The cystic artery is usually less than 3 mm in diameter, and a larger artery in the hepatobiliary triangle is more likely to be the right branch of the hepatic artery.

The origin of the cystic artery is variable ([Fig. 68.10](#)). The most common variant is an origin from the hepatic artery proper, when it often crosses anterior to the common bile duct or common hepatic duct to reach the gallbladder. Rarely, it may arise from the left hepatic, gastroduodenal, superior pancreaticoduodenal, coeliac, right gastric or superior mesenteric arteries; in these cases, the cystic artery may not traverse the hepatobiliary triangle. The cystic artery frequently bifurcates close to its origin, giving rise to two vessels. Alternatively, an accessory cystic artery may arise from the hepatic artery or one of its branches. There appear to be racial differences in the frequency of these variants ([Saidi et al 2007](#)). Small arterial branches from the parenchyma of segment IV or V of the liver may contribute to the supply of the body of the gallbladder, particularly when it is substantially intrahepatic; these help to protect the gallbladder from ischaemic necrosis if inflammation results in thrombosis of the cystic artery.

The cystic artery also gives rise to multiple fine branches that contribute to the blood supply of the extrahepatic bile ducts.

Ductal arteries

The common bile duct and hepatic ducts are supplied by a fine network of arteries arising from several sources. The common bile duct is frequently supplied by 2–4 small-calibre arteries that form a long, tortuous, anastomotic network along its length; these narrow vessels tend to be concentrated at the 3 and 9 o'clock positions around the circumference of the duct ([Northover and Terblanche 1979](#)). The network is fed by arteries from the right branch of the hepatic artery above, reinforced by fine branches from the cystic artery. Branches from three main arterial sources often supply the duct from below: the posterior superior pancreaticoduodenal artery, the retroduodenal artery (both of which originate from the gastroduodenal artery and cross anterior to the

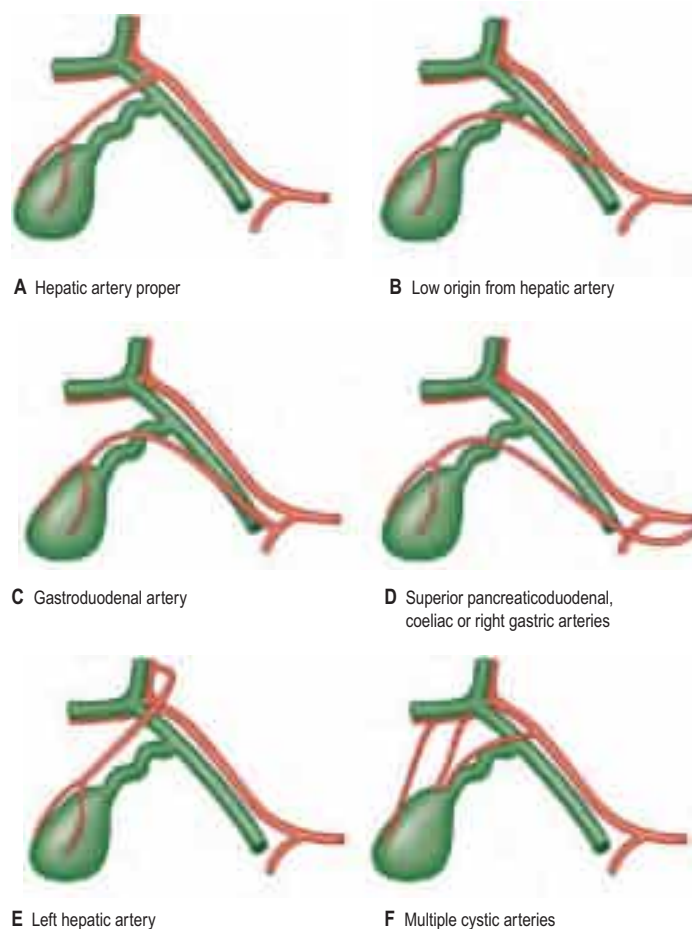


Fig. 68.10 Variant anatomy of the cystic artery.

retroduodenal segment of the bile duct), and a retroportal artery arising from the superior mesenteric artery or coeliac trunk and passing to the right on the posterior surface of the portal vein (Chen et al 1999). The supraduodenal segment of the bile duct is relatively less well vascularized than the retroduodenal and pancreatic segments, and excessive dissection of this segment may result in an ischaemic bile duct stricture.

The hilar ducts are supplied by a fine network of periductal arteries that arise from the right and left branches of the hepatic artery and form an anastomotic plexus within the connective tissue of the hilar plate. The intrahepatic ducts are supplied by segmental branches of the hepatic artery arising within the Glissonian sheaths (Stapleton et al 1998, Vellar 1999). The dependence of the intrahepatic bile ducts on their hepatic arterial blood supply is well recognized in liver transplantation since hepatic artery thrombosis may lead to diffuse intrahepatic biliary strictures despite preservation of portal venous blood flow.

Cystic veins

The venous drainage of the gallbladder is rarely by a single cystic vein. There are usually multiple small veins. Those arising from the superior surface of the body and neck lie in areolar tissue between the gallbladder and liver, and drain into segmental portal veins within the liver. The remainder of the organ drains by one or two small cystic veins into either portal vein branches within the liver or portal venous tributaries draining the hepatic ducts and upper bile duct (Sugita et al 2000). Only rarely does a single or double cystic vein drain directly into the right branch of the portal vein.

Lymphatic drainage

Lymph from the gallbladder and cystic duct drains via several pathways: to the cystic node, which usually lies above the cystic duct in the hepatobiliary triangle, and from here via nodes in the free edge of the lesser omentum and along the common hepatic artery to coeliac lymph nodes; via lymphatics that descend along the common bile duct to the superior retropancreaticoduodenal node (which communicates with para-aortic nodes); and directly to superior mesenteric nodes (Sato et al 2013). Lymphatics on the hepatic aspect of the gallbladder connect directly with intrahepatic lymph vessels.

Lymphatic vessels accompanying the hepatic ducts and upper bile duct drain to hepatic nodes at the porta hepatis and then via lymph nodes in the free edge of the lesser omentum to coeliac nodes. Lymphatics from the lower common bile duct also drain to the latter but some pass directly to retropancreatic and superior mesenteric nodes.

INNERVATION

The gallbladder and the extrahepatic biliary tree are innervated by branches from the hepatic plexus. Gallbladder contraction occurs in response to cholecystokinin (CCK) and parasympathetic (vagal) stimulation. Postganglionic sympathetic nerve fibres from the coeliac and superior mesenteric ganglia are inhibitory to gallbladder smooth muscle. Sympathetic afferents from the gallbladder convey pain sensation; they travel with the greater and lesser splanchnic nerves and have their cell bodies in the T7–9 spinal cord segments. Visceral pain from the gallbladder is referred to the right hypochondrium and epigastrium and may radiate around the back below the right scapula. Inflammation of the parietal peritoneum overlying the gallbladder produces localized right upper quadrant pain. A diverse range of neurotransmitters have been identified within intrinsic neurones of the gallbladder (Balemba et al 2004).

The common bile duct and smooth muscle of the hepatopancreatic ampulla are also innervated by the vagi, either directly or via the hepatic plexus, and by sympathetic nerves.

MICROSTRUCTURE

GALLBLADDER

The fundus of the gallbladder is completely covered by a serosa, and the inferior surfaces and sides of the body and neck of the gallbladder are usually covered by a serosa. If the gallbladder possesses a mesentery, the serosa extends around the sides of the body and neck on to the superior surface and continues into the serosa of the mesentery, whereas the serosa is limited to the inferior surfaces only if the gallbladder is intrahepatic. Beneath the serosa is subserous loose connective and adipose peritoneal tissue. The gallbladder wall microstructure generally

resembles that of the small intestine. The mucosa is yellowish brown and elevated into minute rugae with a honeycomb appearance (see Fig. 68.2). In section, projections of the mucosa into the gallbladder lumen resemble intestinal villi, but they are not fixed structures and the surface flattens as the gallbladder fills with bile. The epithelium is a single layer of columnar cells with apical microvilli; basally, the spaces between epithelial cells are dilated (Fig. 68.11). Many capillaries lie beneath the basement membrane. The epithelial cells actively absorb water and solutes from the bile and concentrate it up to ten-fold. There are no goblet cells in the epithelium. The thin fibromuscular layer is composed of fibrous tissue mixed with smooth muscle cells arranged loosely in longitudinal, circular and oblique bundles.

BILE DUCTS

The walls of the large biliary ducts consist of external fibrous and internal mucosal layers (Fig. 68.12). The outer layer is fibrous connective tissue containing a variable amount of longitudinal, oblique and circular smooth muscle cells. The mucosa is continuous with that in the



Fig. 68.11 A low-power micrograph showing the gallbladder wall, with a mucosal projection that flattens in the full gallbladder, and the thin muscular layer.

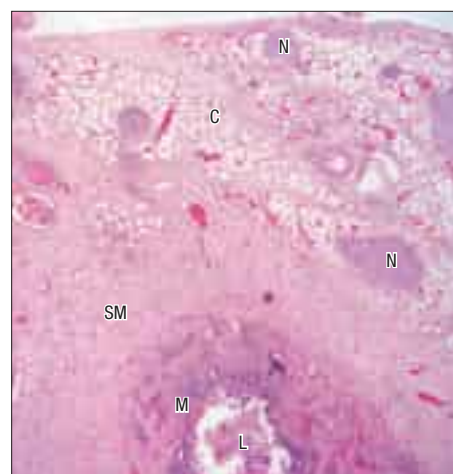


Fig. 68.12 A low-power micrograph showing the structure of a human common bile duct. Shown are the lumen (L), mucosal lining (M), smooth muscle coat (SM) and outer connective tissue (C), with numerous nerve fibre bundles (N). (Courtesy of Mr Peter Helliwell and the late Dr Joseph Mathew, Department of Histopathology, Royal Cornwall Hospitals Trust, UK.)

hepatic ducts, gallbladder and duodenum. The epithelium is columnar and there are numerous tubulo-alveolar (or tubulo-acinar) mucous glands in the duct walls.

Expulsion of gallbladder contents is under neuroendocrine control. Fat in the duodenum causes the release of CCK by intestinal neuroendocrine cells, which stimulates the gallbladder to contract because muscle cells in its walls bear surface receptors for CCK. When the pressure exceeds 100 mmH₂O of bile, the sphincter of Oddi relaxes and bile enters the duodenum. The termination of the united bile and pancreatic ducts is packed with villous, valvular folds of mucosa that contain muscle cells in their connective tissue cores. Contraction is thought to result in retraction and clumping of these folds, so preventing reflux of duodenal contents and controlling the exit of bile.



Bonus e-book images

Fig. 68.3 Hjortsö's crook.

Fig. 68.5 B, Endoscopic image of the major duodenal papilla (arrow) on the medial wall of the descending duodenum.

Fig. 68.7 A scanning electron micrograph showing mucosal folds lining the wall the hepatopancreatic ampulla.

Fig. 68.8 A view of the hepatobiliary triangle at laparoscopic cholecystectomy.

KEY REFERENCES

- Cucchetti A, Peri E, Cescon M et al 2011 Anatomic variations of intrahepatic bile ducts in a European series and meta-analysis of the literature. *J Gastrointest Surg* 15:623–30.
An analysis of 200 cholangiograms from liver donors, coupled with a detailed literature review exploring intrahepatic bile duct anatomy in different racial groups.
- Dasgupta D, Stringer MD 2005 Cystic duct and Heister's 'valves'. *Clin Anat* 18:81–7.
A review of cystic duct anatomy and the significance of its spiral mucosal folds.
- Gross R 1936 Congenital anomalies of gallbladder. A review of one hundred and forty eight cases with a report of double gall-bladder. *Arch Surg* 32:131–62.
A seminal paper on congenital anomalies of the gallbladder.
- Lamah M, Karanjia ND, Dickson GH 2001 Anatomical variations of the extrahepatic biliary tree: review of the world literature. *Clin Anat* 14:167–72.
A literature review of anatomical variations affecting the extrahepatic bile ducts.
- Stringer MD 2009 *Eponyms in Surgery and Anatomy of the Liver, Bile Ducts and Pancreas*. London: Royal Society of Medicine Press.
A short reference book that contains illustrated accounts of the original descriptions of eponyms cited in this chapter, along with a brief biography of the person after whom the structure is named.
- Suzuki M, Akaishi S, Rikiyama T et al 2000 Laparoscopic cholecystectomy, Calot's triangle, and variations in cystic arterial supply. *Surg Endosc* 14:141–4.
An article that documents a consecutive series of 244 laparoscopic cholecystectomies, in which the cystic artery was not present in the hepatobiliary triangle in 11%, and a further 7% of patients had an accessory cystic artery or other arterial variants.
- Talpur KA, Laghari AA, Yousfani SA et al 2010 Anatomical variations and congenital anomalies of extrahepatic biliary system encountered during laparoscopic cholecystectomy. *J Pak Med Assoc* 60:89–93.
An analysis of anatomical variants encountered in 300 adults undergoing cholecystectomy for gallstone disease. Variants were recorded in 20% of patients and predominantly involved the cystic artery (11%), cystic duct (4%), right hepatic artery (3%) and gallbladder (2%).

REFERENCES

- Ashaolu JO, Ukwenya VO, Adenowo TK 2011 Cystoduodenal ligament as an abnormal fold and the accompanying anatomical and clinical implications. *Surg Radiol Anat* 33:171–4.
- Balemba OB, Salter MJ, Mawe GM 2004 Innervation of the extrahepatic biliary tract. *Anat Rec A Discov Mol Cell Evol Biol* 280:836–47.
- Bertelli E, Di Gregorio F, Bertelli L et al 1996 The arterial blood supply of the pancreas: a review. II. The posterior superior pancreaticoduodenal artery. An anatomical and radiological study. *Surg Radiol Anat* 18:1–9.
- Boyden EA 1957 The anatomy of the choledochoduodenal junction in man. *Surg Gynecol Obstet* 104:640–52.
- Castorina S, Scilletta R, Domergue J 2014 Gallbladder agenesis: laparoscopic views of a significant diagnostic challenge. *Surg Radiol Anat* 36: 619–20.
- Chaib E, Kanas AF, Galvão FH et al 2014 Bile duct confluence: anatomic variations and its classification. *Surg Radiol Anat* 36:105–9.
- Chen WJ, Ying DJ, Liu ZJ et al 1999 Analysis of the arterial supply of the extrahepatic bile ducts and its clinical significance. *Clin Anat* 12:245–9.
- Chowbey PK, Wadhwa A, Sharma A et al 2004 Ectopic gallbladder: laparoscopic cholecystectomy. *Surg Laparosc Endosc Percutan Tech* 14:26–8.
- Cucchetti A, Peri E, Cescon M et al 2011 Anatomic variations of intrahepatic bile ducts in a European series and meta-analysis of the literature. *J Gastrointest Surg* 15:623–30.
An analysis of 200 cholangiograms from liver donors, coupled with a detailed literature review exploring intrahepatic bile duct anatomy in different ethnic groups.
- Dahmane R, Morjane A, Starc A 2013 Anatomy and surgical relevance of Rouviere's sulcus. *Scientific World Journal* 2013:254287. doi: 10.1155/2013/254287.
- Dasgupta D, Stringer MD 2005 Cystic duct and Heister's 'valves'. *Clin Anat* 18:81–7.
A review of cystic duct anatomy and the significance of its spiral mucosal folds.
- Di Ciaula A, Wang DQ, Portincasa P 2012 Gallbladder and gastric motility in obese newborns, pre-adolescents and adults. *J Gastroenterol Hepatol* 27:1298–305.
- Didio JA, Anderson MC 1968 Biliary and Pancreatic Ducts: The 'Sphincters' of the Digestive System. Baltimore: Williams & Wilkins, pp. 129–51.
- Faure JP, Doucet C, Scepi M et al 2008 Abnormalities of the gallbladder, clinical effects. *Surg Radiol Anat* 30:285–90.
- Flati G, Flati D, Porowska B et al 1994 Surgical anatomy of the papilla of Vater and the biliopancreatic ducts. *Am Surg* 60:712–18.
- Gross R 1936 Congenital anomalies of gallbladder. A review of one hundred and forty eight cases with a report of double gall-bladder. *Arch Surg* 32:131–62.
A seminal paper on congenital anomalies of the gallbladder.
- Guelrud M, Mendoza S, Rossiter G et al 1990 Sphincter of Oddi manometry in healthy volunteers. *Dig Dis Sci* 35:38–46.
- Guiteau JJ, Fisher M, Cotton RT et al 2009 Intrahepatic gallbladder. *J Am Coll Surg* 209:672.
- Gupta V, Singh V, Sewkani A et al 2009 Torsion of gall bladder, a rare entity: a case report and review article. *Cases J* 2:193.
- Habib FI, Corazzari E, Biliotti D et al 1988 Manometric measurement of human sphincter of Oddi length. *Gut* 29:121–5.
- Kamisawa T, Okamoto A 2008 Pancreatographic investigation of pancreatic duct system and pancreaticobiliary malformation. *J Anat* 212:125–34.
- Kimura K, Ohto M, Saisho H et al 1985 Association of gallbladder carcinoma and anomalous pancreaticobiliary ductal union. *Gastroenterol* 89:1258–65.
- Ko K, Kamiya J, Nagino M et al 2006 A study of the subvesical bile duct (duct of Luschka) in resected liver specimens. *World J Surg* 30:1316–20.
- Lamah M, Karanjia ND, Dickson GH 2001 Anatomical variations of the extrahepatic biliary tree: review of the world literature. *Clin Anat* 14:167–72.
A literature review of anatomical variations affecting the extrahepatic bile ducts.
- Losanoff JE, Jones JW, Richman BW et al 2002 Hepaticocystic duct: a rare anomaly of the extrahepatic biliary system. *Clin Anat* 15:314–15.
- Mirjalili SA, Stringer MD 2011 The arterial supply of the major duodenal papilla and its relevance to endoscopic sphincterotomy. *Endoscopy* 43:307–11.
- Misra SP, Dwivedi M 1990 Pancreaticobiliary ductal union. *Gut* 31: 1144–9.
- Myburgh JA 1993 The Hepp–Couinaud approach to strictures of the bile ducts. I. Injuries, choledochal cysts, and pancreatitis. *Ann Surg* 218:615–20.
- Northover JMA, Terblanche J 1979 A new look at the arterial supply of the bile duct in man and its surgical implications. *Br J Surg* 66: 379–84.
- Perret RS, Sloop GD, Borne JA 2000 Common bile duct measurements in an elderly population. *J Ultrasound Med* 19:727–30.
- Purvis NS, Mirjalili SA, Stringer MD 2013 The mucosal folds at the pancreaticobiliary junction. *Surg Radiol Anat* 35:943–50.
- Saidi H, Karanja TM, Ogengo JA 2007 Variant anatomy of the cystic artery in adult Kenyans. *Clin Anat* 20:943–5.
- Sakai Y, Tsuyuguchi T, Sugiyama H et al 2012 Current situation of endoscopic treatment for common bile duct stones. *Hepatogastroenterol* 59:1712–16.
- Sato T, Ito M, Sakamoto H 2013 Pictorial dissection review of the lymphatic pathways from the gallbladder to the abdominal para-aortic lymph nodes and their relationships to the surrounding structures. *Surg Radiol Anat* 35:615–21.
- Shaw MJ, Dorsher PJ, Vennes JA 1993 Cystic duct anatomy: an endoscopic perspective. *Am J Gastroenterol* 88:2102–6.
- Singh B, Ramsaroop L, Allopi L et al 2006 Duplicate gallbladder: an unusual case report. *Surg Radiol Anat* 28:654–7.
- Spanos CP, Syrakos T 2006 Bile leaks from the duct of Luschka (subvesical duct): a review. *Langenbecks Arch Surg* 391:441–7.
- Stapleton GN, Hickman R, Terblanche J 1998 Blood supply of the right and left hepatic ducts. *Br J Surg* 85:202–7.
- Stringer MD 2009 Eponyms in Surgery and Anatomy of the Liver, Bile Ducts and Pancreas. London: Royal Society of Medicine Press.
A short reference book that contains illustrated accounts of the original descriptions of eponyms cited in this chapter, along with a brief biography of the person after whom the structure is named.
- Stringer MD, Dhawan A, Davenport M et al 1995 Choledochal cysts: lessons from a 20 year experience. *Arch Dis Child* 73:528–31.
- Sugita M, Ryu M, Satake M et al 2000 Intrahepatic inflow areas of the drainage vein of the gallbladder: analysis by angio-CT. *Surgery* 128: 417–21.
- Suzuki M, Akaishi S, Rikiyama T et al 2000 Laparoscopic cholecystectomy, Calot's triangle, and variations in cystic arterial supply. *Surg Endosc* 14:141–4.
An article that documents a consecutive series of 244 laparoscopic cholecystectomies, in which the cystic artery was not present in the hepatobiliary triangle in 11%, and a further 7% of patients had an accessory cystic artery or other arterial variants.
- Talpur KA, Laghari AA, Yousfani SA et al 2010 Anatomical variations and congenital anomalies of extrahepatic biliary system encountered during laparoscopic cholecystectomy. *J Pak Med Assoc* 60:89–93.
An analysis of anatomical variants encountered in 300 adults undergoing cholecystectomy for gallstone disease. Variants were recorded in 20% of patients and predominantly involved the cystic artery (11%), cystic duct (4%), right hepatic artery (3%) and gallbladder (2%). Anatomical variants were associated with a slightly increased incidence of postoperative morbidity.
- Teilum D 1991 In vitro measurement of the length of the sphincter of Oddi. *Endoscopy* 23:114–6.
- Vellar ID 1999 The blood supply of the biliary ductal system and its relevance to vasculobiliary injuries following cholecystectomy. *Aust N Z J Surg* 69:816–20.
- Woods CM, Mawe GM, Toouli J et al 2005 The sphincter of Oddi: understanding its control and function. *Neurogastroenterol Motil* 17:31–40.